



**On the Optimality of the Generalized Power Method for SNR
Maximization via RIS Phase Shift Design**

Journal:	<i>IEEE Transactions on Wireless Communications</i>
Manuscript ID	Paper-TW-Oct-25-2910
Manuscript Type:	Original Transactions Paper
Date Submitted by the Author:	06-Nov-2025
Complete List of Authors:	Khemiri, Moez; King Abdullah University of Science and Technology Ye, Jia; Chongqing University Kammoun, Abla; King Abdullah University of Science and Technology Alouini, Mohamed; King Abdullah University of Science and Technology
Keyword:	

SCHOLARONE™
Manuscripts

On the Optimality of the Generalized Power Method for SNR Maximization via RIS Phase Shift Design

Moez Khemiri, Jia Ye, *Member, IEEE*, Abba Kammoun, *Member, IEEE*, and Mohamed-Slim Alouini, *Fellow, IEEE*

Abstract—Reconfigurable intelligent surfaces (RIS) phase constraints present a significant challenge for realizing feasible and efficient RIS network design. In this work, a unified framework based on gradient projection is presented as a generalized power method (GPM) for the quadratic signal to noise ratio (SNR) maximization problem. Increasingly realistic phase constraints are imposed, progressing from an idealized continuous unit-modulus model to a more practical phase-amplitude coupled model. For each constraint, the GPM updates are efficiently derived, transforming the quadratic problem into a linear projection step. Karush–Kuhn–Tucker (KKT) analysis shows that under continuous unit-modulus, discrete unit-modulus, and phase-coupled continuous constraints, the algorithm converges to a point verifying the stationarity condition. A minimum achievable SNR scaling is derived through the decomposition of the objective, which quantifies the impact of these constraints. Numerical results demonstrate that the GPM framework is well-suited to the rank-deficient nature of the quadratic SNR maximization problem, given that it is faster than other gradient methods, outperforms greedy methods, and maintains linear complexity across all constraints.

Index Terms—Reconfigurable intelligent surfaces, quadratic optimization, power method, SNR maximization, practical phase shifts

I. INTRODUCTION

The maximization problem over a compact set is the most common optimization problem in reconfigurable intelligent surfaces (RIS) assisted wireless network design, whose objective can be SNR, signal to interference noise ratio (SINR), capacity, sum rate, energy efficiency, etc. By representing the objective as $f(\phi)$ and the feasible set as $\mathcal{Q}$, the maximization problem is expressed as

$$f^* = \max_{\phi \in \mathcal{Q}} f(\phi). \quad (1)$$

The set $\mathcal{Q}$ of the RIS reflecting vectors ϕ is typically non-convex as it can be the ideal continuous unit-modulus set [1], a discrete phase shift set [2], or more realistically, a discrete phase-dependent amplitude set [3]. These choices reflect a progression from idealized to hardware-constrained models, each with algorithmic and theoretical implications that become more challenging as the number of reflecting vectors increases for large-scale multi-input-multi-output (MIMO) systems.

M. Khemiri, A. Kammoun, and M.-S. Alouini are with the Computer, Electrical and Mathematical Sciences & Engineering (CEMSE) Division, King Abdullah University of Science and Technology (KAUST), Thuwal 23955-6900, Saudi Arabia (email: moez.khemiri@kaust.edu.sa; abba.kammoun@kaust.edu.sa; slim.alouini@kaust.edu.sa).

J. Ye is with the School of Electrical Engineering, Chongqing University, Chongqing 400044, China (email: jia.ye@cqu.edu.cn).

Corresponding author: Jia Ye (email: jia.ye@cqu.edu.cn).

Despite the prevalence of these problems, their theoretical understanding remains limited as we impose more constraints. When the feasible set involves discrete phase shifts, Eq. (1) becomes a challenging non-convex combinatorial task. Exact methods, such as exhaustive [4], [5] or branch-and-bound search [6], [7], are impractical for large-scale RIS configurations due to their exponential complexity. To address this, approximate techniques have been explored, including greedy heuristics [8], [9], local search [10], and learning-based strategies [11]. While these approaches are computationally efficient and often perform well, they typically lack theoretical guarantees. Moreover, due to the channel-dependent nature of the optimization objective, the performance and convergence of heuristic methods often vary across realizations.

Beyond algorithmic complexity, hardware constraints play a significant role in shaping the problem's difficulty. In [12], practical challenges are highlighted, notably that each RIS element is limited to discrete phase control. Moreover, inter-element mutual coupling induces correlation effects, further complicating the design. Real-time implementation is another bottleneck, as optimizations must be completed within the channel coherence time [13] (a few milliseconds). This necessitates specialized hardware designs using fixed-point arithmetic, pipelined processing, and efficient algorithms, as demonstrated in field-programmable gate array (FPGA)-based prototypes.

The number of reflecting phase shift elements has increased, as large RIS arrays are essential to achieve meaningful beam-forming gains and spatial diversity. For example, the RIS-based IMT-2030 test-bed [14] features $N = 1024$ passive phase shift elements, and $M = 64$ base station antennas operating at mmWave frequencies. As RIS deployments scale up, this gives ground to the development of scalable, low-complexity algorithms that offer empirical performance and theoretical guarantees under realistic hardware and channel constraints. In this work, we propose a unified projected gradient framework tailored to compact sets and demonstrate its effectiveness across continuous, discrete, and amplitude-phase configurations for SNR maximization.

A. Related works

The RIS phase shift optimization constraints have been studied using various methods, particularly for the continuous unit modulus constraint, for which gradient methods can be applied with slight adjustments. Quadratic objectives under unit modulus constraints have been solved by unimodular quadratic

programming (UQP). The study in [15] lays a foundation for UQP by analyzing the convergence properties of a power method-like iterative scheme. They prove the update converges locally to a stable point and introduce a monotonically error-bound improving technique to address the UQP in radar code design. Projected gradient descent was applied in [16] to the unit-modulus least squares problem, using element-wise phase normalization after each step. Using real variables, the authors derive Karush–Kuhn–Tucker (KKT) conditions and establish a linear convergence rate typical of gradient-based methods. Similarly, [17] optimizes the continuous phase shifts via a fractional programming-based iterative update resembling gradient refinement. A quadratic-quartic optimization problem for beyond-diagonal RIS was addressed in [18] for integrated sensing and communication (ISAC). Their solution is a penalty-based alternating optimization method handling continuous and discrete constraints using a power-method-like update.

Discrete methods require more nuanced algorithmic approaches than their continuous counterparts. For example, a relaxation method was proposed in [19], where the gradient update iteratively transitions from unconstrained to discrete phases, which was shown to improve performance compared to enforcing the discrete constraints. A method presented in [20] addresses single-input-single-output (SISO) RIS optimization for discrete-phase constraints. The authors derive a closed-form optimality condition for element-wise phase updates. Their algorithm converges in at most N steps, sequentially updating each RIS element to align its phase with the incident signals, ensuring constructive interference. Multiple-input-single-output (MISO) communications are addressed in [21], where the authors propose a low-complexity RIS element allocation algorithm for orthogonal frequency-division multiplexing (OFDM) systems. The algorithm allocates each RIS element to a single sub-band, then optimizes it using a closed-form expression designed to maximize its sub-band channel gain. For MIMO, a linear method was proposed in [22], where discrete phase optimization is obtained via iterative closed-form update using linear operators derived from channel estimation.

Amplitude losses due to inter-element coupling are modeled as amplitude-phase coupling. A greedy method to address this has been proposed in [9], which formulates a series of element-wise linear problems to update each phase. Suboptimal optimization under practical constraints was also proposed in [23], where the optimization alternates between shifts and the transmit covariance matrix. A joint RIS optimization and power allocation algorithm was presented in [24] using the Newton-Raphson method to optimize the phase-amplitude coupled RIS elements.

The channel-dependent nature of the objective and its effect on SNR scaling are crucial considerations. The paper by [25] challenges the far-field assumption, showing that the conventional $O(N^2)$ SNR scaling for an RIS-MISO system saturates in the near-field, providing a realistic performance limit. Separately, the work in [4] shows that sufficient quantization in a discrete-phase RIS can match ideal continuous-phase SNR for rank-1 channels.

Despite research into individual constraints, there is no unified algorithmic framework to handle all RIS phase shift problems, especially the less-studied discrete phase-amplitude coupling, in terms of computational complexity and performance. Additionally, the existing SNR scaling laws offer limited insight into the performance of the solutions under random channel distributions.

B. Main contributions

In this work, we present tools for optimization in RIS-assisted systems, summarized by the following contributions:

- We present a gradient scheme proposed in [26, Alg. 1] based on the first-order Taylor expansion for any convex function over a compact set. When the objective is quadratic, as in SNR maximization, the gradient update set transforms the problem into a linear optimization task referred to as the generalized power method (GPM).
- We apply GPM to continuous and discrete unit-modulus, and amplitude-phase coupled RIS configurations. For the discrete unit-modulus case, we introduce a projection with variable-resolution refinement to improve convergence for low-bit phase quantization. For amplitude-phase coupling, we prove that the projection reduces to a linear maximization, which we simplify using the amplitude function, resulting in algorithms with linear complexity for both continuous and discrete settings.
- Using Wirtinger’s calculus to simplify complex derivatives, we prove that the limit point satisfies KKT stationarity conditions. We also utilize real-valued constraints with complex variables for discrete and continuous problems. For amplitude-phase coupling, we utilize the projected phase condition to derive stationarity conditions, proving that the algorithm converges to a stationary point under realistic constraints.
- By leveraging the low-rank structure of the cascaded channel and monotonicity of GPM iterates, we construct a lower bound on the ergodic SNR by aligning with the strongest path component. This provides insights into the expected $O(N^2M)$ scaling and characterizes the effect of quantization and amplitude-phase coupling.
- We provide comparisons with existing works on complexity, required iterations, and runtime, proving that under amplitude-phase coupling and discrete phase shifts, we maintain linear complexity with optimal performance.

C. Notations and outline

In this paper, x denotes a scalar, $\mathbf{x}$ represents a vector, and $\mathbf{X}$ stands for a matrix. x_n denotes the n th entry of $\mathbf{x}$. For complex-valued quantities, $(\cdot)^H$ is the conjugate transpose, $(\cdot)^T$ denotes the transpose, and $(\cdot)^*$ represents the conjugate. Moreover, $\text{diag}(\mathbf{x})$ is a diagonal matrix whose diagonal entries are taken from the vector $\mathbf{x}$. The Hadamard (element-wise) product of two vectors $\mathbf{x}, \mathbf{y} \in \mathbb{C}^N$ is denoted by $\mathbf{x} \circ \mathbf{y}$, where $(\mathbf{x} \circ \mathbf{y})_n = x_n y_n$ for all n . The argument of a complex number x , denoted by $\arg(x)$, represents the phase in the principal range $(-\pi, \pi]$, with the convention that $\arg(0) = 0$. The Euclidean (or ℓ_2) norm of a vector $\mathbf{x}$ is represented by $\|\mathbf{x}\|_2$,

similarly for the ℓ_1 norm. The floor function, denoted as $\lfloor x \rfloor$, returns the greatest integer less than or equal to x . Big O notation, written as $a = O(b)$, serves as an asymptotic upper bound such that $|a| \leq cb$ for some constant c .

The rest of this paper is structured as follows: Section II defines the RIS SNR maximization problem as a quadratic function of the channel correlation matrix and the phase shift vector, providing the background for the generalized power method for complex quadratic optimization problems. Sections III and IV describe the solutions for the unit modulus and practical phase-amplitude problems, respectively. Section V derives an ergodic lower bound on the achievable SNR. Section VI presents an analysis of the proposed algorithms and complexity comparisons with existing works before providing numerical results in Section VII. Finally, Section VIII provides conclusions for this paper.

II. PROBLEM OVERVIEW

This section introduces the system model for an RIS-assisted wireless system and formulates the SNR maximization problem. The problem is shown to be the maximization of a convex quadratic objective over a compact set. We then introduce a general iterative framework to solve this problem, specialize it to quadratic objectives, and prove its convergence.

A. RIS SNR maximization problem

We consider an RIS-assisted wireless system where a base station (BS) equipped with multiple antennas communicates with a single-antenna user (UE). The downlink signal at the user is given by

$$y_d = \sqrt{P_d} \mathbf{h}^H \Phi \mathbf{G} \mathbf{w}_d x_d + \xi_d, \quad (2)$$

where P_d is the BS transmit power, $\mathbf{h} \in \mathbb{C}^N$ and $\mathbf{G} \in \mathbb{C}^{N \times M}$ represent the RIS-user and BS-RIS channels, respectively. The RIS applies phase shifts via $\Phi = \text{diag}(\phi)$, with $\phi = [\phi_1, \dots, \phi_N]^T$. The BS transmits using a precoding vector $\mathbf{w}_d$, while x_d is the unit-power transmit symbol. The received signal is corrupted by $\xi_d \sim \mathcal{CN}(0, \sigma_d)$. The BS-RIS channel $\mathbf{G}$ is modeled as

$$\mathbf{G} = \sqrt{\frac{NM}{N_G}} \sum_{p=1}^{N_G} \alpha_p \mathbf{a}_N(\theta_{\text{RIS}_p}) \mathbf{a}_M^H(\theta_{\text{BS}_p}), \quad (3)$$

where N_G denotes the number of propagation paths and α_p is the complex gain of the p th path. The array steering vector of size $X \in \{M, N\}$, for half-wavelength spacing, is defined as $\mathbf{a}_X(\theta) = \frac{1}{\sqrt{X}} [1, e^{j\pi \sin(\theta)}, \dots, e^{j(X-1)\pi \sin(\theta)}]^T$. Here, θ_{RIS_p} represents the angle-of-arrival (AoA) at the RIS from the BS, and θ_{BS_p} represents the angle-of-departure (AoD) from the BS towards the RIS. The RIS-user channel vector $\mathbf{h}$ is given by

$$\mathbf{h} = \sqrt{\frac{N}{N_h}} \sum_{i=1}^{N_h} \beta_i \mathbf{a}_N(\theta_{\text{UE}_i}), \quad (4)$$

where N_h denotes the number of propagation paths in the RIS-user link and β_i is the complex gain of the i th path

with corresponding angle-of-arrival θ_{UE_i} at the RIS. The instantaneous user SNR is

$$\text{SNR} = \frac{P_d}{\sigma_d} |\mathbf{h}^H \Phi \mathbf{G} \mathbf{w}_d|^2. \quad (5)$$

Using maximum ratio transmission, the optimal downlink precoder is $\mathbf{w}_d = \mathbf{G}^H \Phi \mathbf{h} / \|\mathbf{G}^H \Phi \mathbf{h}\|_2$. Substituting this, the problem becomes

$$\max_{\phi \in \mathcal{Q}} \phi^T \text{diag}(\mathbf{h}^*) \mathbf{G} \mathbf{G}^H \text{diag}(\mathbf{h}) \phi^*. \quad (6)$$

Under a time division duplex (TDD) system, channel reciprocity implies that uplink and downlink channels are equivalent, facilitating both channel estimation and beamforming design. Recent works on RIS-aided channel estimation, such as pilot-based and compressed sensing techniques, enable accurate acquisition of the effective cascaded channel [27]. The uplink signal received at the BS is modeled as

$$y_u = \sqrt{P_u} \mathbf{w}_u^H (\mathbf{G}^H \Phi \mathbf{h}) s_u + \xi_u, \quad (7)$$

where P_u is the transmit power, s_u is a unit-power symbol, and $\xi_u \sim \mathcal{CN}(0, \sigma_u)$. Given that RIS elements are passive, only the cascaded channel $\mathbf{G}^H \text{diag}(\phi^*) \mathbf{h}$ can be estimated. Using the identity

$$\mathbf{G}^H \Phi \mathbf{h} = \mathbf{G}^H \text{diag}(\phi^*) \mathbf{h} = (\phi^H \text{diag}(\mathbf{h}) \mathbf{G}^*)^T, \quad (8)$$

and defining the cascaded channel matrix as $\mathbf{H} = \text{diag}(\mathbf{h}) \mathbf{G}^*$, the SNR maximization problem in Eq. (6) can be equivalently written as

$$\max_{\phi \in \mathcal{Q}} \phi^T \mathbf{H}^* \mathbf{H}^T \phi^* = \max_{\phi \in \mathcal{Q}} \phi^H \mathbf{H} \mathbf{H}^H \phi = \max_{\phi \in \mathcal{Q}} f(\phi), \quad (9)$$

where $\text{rank}(\mathbf{H}) = \text{rank}(\mathbf{G}) = N_G$. In this work, we assume that $\mathbf{H}$ is perfectly known and solve the resulting quadratic problem under phase shift constraints described by the set $\mathcal{Q}$.

B. Proposed framework

To solve the SNR maximization problem in Eq. (9), we employ an iterative gradient-based scheme.

1) *Gradient scheme algorithm*: The gradient scheme algorithm, described in [26], is an iterative optimization technique based on the first-order Taylor expansion of the objective function. We denote the k th iterate as $\phi^{(k)}$ and the limit point as ϕ^* .

Algorithm 1 Gradient scheme [26, Alg. 1]

- 1: **Input:** Initial vector $\phi^{(0)} \in \mathcal{Q}$
 - 2: **for** $k = 0$ to K **do**
 - 3: $\phi^{(k+1)} = \text{Arg max}_{\phi \in \mathcal{Q}} f(\phi^{(k)}) + \langle \nabla f(\phi^{(k)}), \phi - \phi^{(k)} \rangle$
 - 4: **end for**
-

Given a function $f(\phi)$, its local approximation around $\phi^{(k)}$ is $f(\phi) \approx f(\phi^{(k)}) + \langle \nabla f(\phi^{(k)}), \phi - \phi^{(k)} \rangle$. At each iteration, the update is obtained by maximizing this approximation over the compact set $\mathcal{Q}$ as

$$\phi^{(k+1)} = \text{Arg max}_{\phi \in \mathcal{Q}} \phi^{(k)} + \langle \nabla f(\phi^{(k)}), \phi - \phi^{(k)} \rangle. \quad (10)$$

Since both $f(\phi^{(k)})$ and $\langle \nabla f(\phi^{(k)}), \phi^{(k)} \rangle$ are constant with respect to ϕ , the maximization reduces to

$$\phi^{(k+1)} = \text{Arg max}_{\phi \in \mathcal{Q}} \langle \nabla f(\phi^{(k)}), \phi \rangle. \quad (11)$$

Definition 1. The projection of a vector $\mathbf{x}$ onto $\mathcal{Q}$ is given by

$$\text{Proj}_{\mathcal{Q}}(\mathbf{x}) = \text{Arg max}_{\phi \in \mathcal{Q}} \langle \mathbf{x}, \phi \rangle.$$

The original problem is transformed into maximizing the dot product $\text{Re}(\mathbf{x}^H \phi)$, where $\mathbf{x} = \nabla f(\phi)$. This reformulation reduces the complexity of the problem and transforms it into a linear step. By iteratively applying this step, the method ensures that ϕ remains feasible while improving the objective value, as will be formally shown in the next section.

2) *Proof of convergence:* For the sake of completeness, we revisit the proof found in [26, Theorem 1] to establish the convergence. We define the optimality gap as

$$\Delta(\phi^{(k)}) = \max_{\phi \in \mathcal{Q}} \langle \nabla f(\phi^{(k)}), \phi - \phi^{(k)} \rangle. \quad (12)$$

By convexity of f , we have $\Delta(\phi^{(k)}) \geq 0$, with equality holding when the gradient $\nabla f(\phi^{(k)})$ belongs to the normal cone of $\text{Conv}(\mathcal{Q})$ at $\phi^{(k)}$. Using convexity on the first-order Taylor expansion, we obtain

$$\begin{aligned} f(\phi^{(k+1)}) &\geq f(\phi^{(k)}) + \langle \nabla f(\phi^{(k)}), \phi^{(k+1)} - \phi^{(k)} \rangle \\ &= f(\phi^{(k)}) + \Delta(\phi^{(k)}) \geq f(\phi^{(k)}). \end{aligned} \quad (13)$$

Since $\Delta(\phi^{(k)}) \geq 0$, we conclude that the sequence $\{f(\phi^{(k)})\}_{k=0}^{\infty}$ is monotonically increasing. Summing over K iterations, we obtain

$$f(\phi^*) - f(\phi^{(0)}) \geq f(\phi^{(K+1)}) - f(\phi^{(0)}) \geq \sum_{k=0}^K \Delta(\phi^{(k)}). \quad (14)$$

Defining the cumulative improvement as $S^{(K)} = \sum_{k=0}^K \Delta(\phi^{(k)})$, we note that $S^{(K)}$ is an increasing and bounded sequence. By the monotone convergence theorem, $S^{(K)}$ converges, and we conclude that $\lim_{K \rightarrow \infty} S^{(K)} = \sum_{k=0}^{\infty} \Delta(\phi^{(k)})$. Since the series is convergent, the general term must satisfy $\lim_{k \rightarrow \infty} \Delta(\phi^{(k)}) = 0$. This confirms that the optimality gap goes to zero and that the algorithm converges to a stationary point.

3) *Generalized power method (GPM):* Our SNR maximization problem is a specific instance of the general problem, where the objective is the quadratic $f(\phi) = \phi^H \mathbf{A} \phi$ with $\mathbf{A} = \mathbf{H} \mathbf{H}^H$. The gradient is thus $\nabla f(\phi^{(k)}) = \mathbf{A} \phi^{(k)}$. Substituting this into the update rule from Eq. (11), the projection is given by

$$\phi^{(k+1)} = \text{Proj}_{\mathcal{Q}}(\mathbf{A} \phi^{(k)}). \quad (15)$$

We refer to this framework as the generalized power method or GPM, similar to the classical power method where the projection step is given as $\phi^{(k+1)} = \mathbf{A} \phi^{(k)} / \|\mathbf{A} \phi^{(k)}\|_2$. The GPM generalizes this concept to any compact set $\mathcal{Q}$ by transforming the quadratic optimization problem into an iterative linear maximization problem.

III. UNIT MODULUS PHASE ELEMENTS

In this section, we apply GPM to the quadratic SNR maximization problem using unit modulus phase elements with both continuous and discrete phase shifts. We derive the projection step and KKT conditions explicitly, and introduce a variable resolution schedule to improve convergence in low-bit resolutions.

A. Continuous phase RIS problem

We first use GPM for unit-modulus phase elements with continuously adjustable phase shifts.

1) *GPM for continuous phase optimization:* Considering the ideal RIS with continuously adjustable phase shifts without any amplitude loss or gain, the compact set is given as $\mathcal{Q} = \mathcal{F}_{\infty}^N = \{\phi \in \mathbb{C}^N \mid |\phi_n| = 1, \forall n \in \{1, \dots, N\}\}$. In this case, the projection step in Eq. (15) is transferred to

$$\begin{aligned} \phi^{(k+1)} &= \text{Proj}_{\mathcal{F}_{\infty}^N}(\mathbf{H} \mathbf{H}^H \phi^{(k)}) \\ &= \exp j \arg(\mathbf{H} \mathbf{H}^H \phi^{(k)}), \end{aligned} \quad (16)$$

where $\exp j \arg(\cdot)$ is applied element-wise. Equivalently we write

$$\text{Proj}_{\mathcal{F}_{\infty}^N}(\mathbf{H} \mathbf{H}^H \phi^{(k)}) = \left[\frac{(\mathbf{H} \mathbf{H}^H \phi^{(k)})_1}{|(\mathbf{H} \mathbf{H}^H \phi^{(k)})_1|}, \dots, \frac{(\mathbf{H} \mathbf{H}^H \phi^{(k)})_N}{|(\mathbf{H} \mathbf{H}^H \phi^{(k)})_N|} \right]^T. \quad (17)$$

The continuous GPM algorithm involves two main steps: computing and projecting the gradient. The projection step enforces the unit modulus constraint with complexity $O(N)$. To avoid computing the full correlation matrix $\mathbf{H} \mathbf{H}^H$, we instead compute the gradient efficiently by first evaluating $\mathbf{H}^H \phi^{(k)}$, followed by multiplying the result with $\mathbf{H}$. This reduces the complexity from $O(N^2)$ to $O(NM)$, which is especially beneficial in large-scale scenarios where $N \gg M$. By iteratively computing and projecting the gradient, we improve the objective function until a convergence criterion is met. For the unit ball problem, the global solution corresponds to the dominant eigenvector of $\mathbf{H} \mathbf{H}^H$. However, when a unit modulus constraint is imposed, the global optimum is not generally achievable, given the non-convex constraints. Some conditions on unit modulus global optima are given in [15], particularly for rank-1 matrices ($N_G = 1$). In our general case, let the compact eigenvalue decomposition (EVD) of $\mathbf{H} \mathbf{H}^H = \mathbf{V}_c \mathbf{\Lambda}_c \mathbf{V}_c^H$, where $\mathbf{\Lambda}_c \in \mathbb{R}^{N_G \times N_G}$ contains the N_G nonzero eigenvalues. The objective can be written as

$$f(\phi) = \phi^H \mathbf{H} \mathbf{H}^H \phi = \phi^H \mathbf{V}_c \mathbf{\Lambda}_c \mathbf{V}_c^H \phi = \sum_{p=1}^{N_G} \lambda_p |\mathbf{v}_p^H \phi|^2, \quad (18)$$

where λ_p is the p th eigenvalue of $\mathbf{H} \mathbf{H}^H$ and $\mathbf{v}_p$ is the corresponding orthonormal eigenvector. The work in [28] proposes low complexity methods to align with the dominant eigenvector within the unit modulus set that show improved performance under some conditions on the matrix. Letting $e^{j \arg(\mathbf{v}_1)}$ be the projection of the dominant eigenvalue λ_1 and using the fact that for unit modulus elements $\|\phi\|_2 = \sqrt{N}$, we have that

$$N \lambda_1 \geq \max_{\phi \in \mathcal{F}_{\infty}^N} f(\phi) \geq \lambda_1 \|\mathbf{v}_1\|_1^2. \quad (19)$$

In the following, we will prove that the limit point of GPM is a fixed point that satisfies KKT conditions.

2) *Unit modulus continuous stationarity condition:* The continuous unit modulus constraint $\phi \in \mathcal{F}_\infty^N$ can be transformed as $|\phi_n| = 1 \iff \sqrt{\phi_n^* \phi_n} = 1, \forall n \in \{1, \dots, N\}$. Let $\mu \in \mathbb{R}^N$, we define the Lagrangian as

$$\mathcal{L}(\phi, \mu) = \phi^H \mathbf{H} \mathbf{H}^H \phi - \sum_{n=1}^N \mu_n \left(\sqrt{\phi_n^* \phi_n} - 1 \right). \quad (20)$$

The stationarity condition is $\partial \mathcal{L} / \partial \phi^* = \mathbf{H} \mathbf{H}^H \phi - \mu \circ \phi / (2|\phi|) = \mathbf{0}$, where the division is applied element-wise. Explicitly, for each component the stationarity condition is

$$(\mathbf{H} \mathbf{H}^H \phi)_n - \mu_n \frac{\phi_n}{2|\phi_n|} = 0, \quad \forall n \in \{1, \dots, N\}. \quad (21)$$

At convergence, the algorithm reaches a fixed point ϕ^* , where $\phi_n^* = (\mathbf{H} \mathbf{H}^H \phi^*)_n / |(\mathbf{H} \mathbf{H}^H \phi^*)_n|$. Let $M_n^* = |(\mathbf{H} \mathbf{H}^H \phi^*)_n|$, then $\mu_n^* = 2|M_n^*| \in \mathbb{R}$ and $|\phi_n^*| = 1, \forall n \in \{1, \dots, N\}$ and verify that

$$\mathbf{H} \mathbf{H}^H \phi^* - \mu^* \circ \frac{\phi^*}{2|\phi^*|} = \mathbf{0}. \quad (22)$$

B. Discrete phase RIS problem

In this section, we employ the GPM to optimize the RIS configuration, where the phase shifts are restricted to a discrete set of allowable values.

1) *GPM for discrete phase optimization:* Considering the ideal RIS with discrete phase shifts without any amplitude loss or gain, the compact set is given as $\mathcal{Q} = \mathcal{F}_b^N = \left\{ \phi \in \mathbb{C}^N \mid \phi_n \in \left\{ \exp(j2\pi m/2^b) \right\}_{m=0}^{2^b-1} \right\}$ with b being the phase resolution in number of bits. The main optimization step is given as

$$\phi^{(k+1)} = \text{Proj}_{\mathcal{F}_b^N} \left(\mathbf{H} \mathbf{H}^H \phi^{(k)} \right). \quad (23)$$

The discrete projection $\text{Proj}_{\mathcal{F}_b^N}(\mathbf{H} \mathbf{H}^H \phi^{(k)})$ is given by matching the closest discrete phase as

$$\left[\exp(j\frac{2\pi}{2^b} \left[\frac{\vartheta_1^{(k)} 2^b}{2\pi} + \frac{1}{2} \right]), \dots, \exp(j\frac{2\pi}{2^b} \left[\frac{\vartheta_N^{(k)} 2^b}{2\pi} + \frac{1}{2} \right]) \right]^T, \quad (24)$$

where $\vartheta_n^{(k)} = \arg((\mathbf{H} \mathbf{H}^H \phi^{(k)})_n) \in (-\pi, \pi]$ is the complex argument of the n th component in the vector $\mathbf{H} \mathbf{H}^H \phi^{(k)}$. Unlike the continuous phase projection, the discrete projection is restricted to finding the closest match within the feasible set, inevitably introducing a quantization error that can accumulate and affect the convergence.

Lemma 1. *The phase quantization error, which measures the difference between the continuous and discrete phase projection, is defined as $E_b(\vartheta_n) = \vartheta_n - 2\pi \lfloor 2^b \vartheta_n / 2\pi + \frac{1}{2} \rfloor / 2^b$. We have then*

$$|E_b(\vartheta_n)| \leq \frac{\pi}{2^b}.$$

The quantization error becomes less significant with a higher phase resolution, which allows early iterations to overcome poor initializations and converge toward a point that approaches the continuous solution. To capture this effect, we

introduce a variable resolution schedule that starts from a near-continuous setting and decays exponentially as

$$b^{(k)} = \lfloor b + (B - b)e^{-\gamma k} \rfloor, \quad \gamma > 0, \quad B \gg 1. \quad (25)$$

Starting with $b^{(0)} = B$, by Lemma 1, we can show that the projection $\text{Proj}_{\mathcal{F}_B^N}(\mathbf{H} \mathbf{H}^H \phi^{(0)})$ approximates the continuous vector $\text{Proj}_{\mathcal{F}_\infty^N}(\mathbf{H} \mathbf{H}^H \phi^{(0)})$ with an error $O(\sqrt{N}2^{-B})$. This ensures that early iterations are close to the continuous trajectory, avoiding early convergence toward suboptimal discrete configurations due to poor initialization. The nested feasible sets $\mathcal{F}_{b^{(k+1)}}^N \subseteq \mathcal{F}_{b^{(k)}}^N$ then gradually enforce coarser phase constraints, starting from the higher resolution, aiming to converge to a better solution in the feasible set.

2) *Unit modulus discrete stationarity condition:* As a first step, we note that the discrete phase constraint can be transformed as in Lemma 2.

Lemma 2. *Let $\phi \in \mathbb{C}^N$, we have that*

$$\phi \in \mathcal{F}_b^N \iff \phi_n^{2^b} = 1, \quad \forall n \in \{1, \dots, N\}.$$

Secondly, we see that the constraint $\mathbf{g}(\phi)$ defined as $\mathbf{g}(\phi)_n = \phi_n^{2^b} - 1, \forall n \in \{1, \dots, N\}$, is a complex-valued function, we then need to express it in terms of its real and imaginary parts $\mathbf{g}_R(\phi) = \text{Re}(\mathbf{g}(\phi))$ and $\mathbf{g}_I(\phi) = \text{Im}(\mathbf{g}(\phi))$. We can finally write the Lagrangian $\mathcal{L}(\phi, \mu_R, \mu_I)$ for $f(\phi)$ subject to $\phi \in \mathcal{F}_b^N$ as

$$\phi^H \mathbf{H} \mathbf{H}^H \phi - \sum_{n=1}^N \mu_{Rn} \text{Re}(\mathbf{g}(\phi)_n) + \mu_{In} \text{Im}(\mathbf{g}(\phi)_n), \quad (26)$$

where $\mu_R, \mu_I \in \mathbb{R}^N$ are the multipliers for the real and imaginary part of the constraint. According to [29, Eq. (148)] the Lagrangian can equivalently be written as

$$\mathcal{L}(\phi, \mu_R, \mu_I) = \phi^H \mathbf{H} \mathbf{H}^H \phi - \text{Re} \left((\mu_R + j\mu_I)^H \mathbf{g}(\phi) \right). \quad (27)$$

The stationarity condition is given as

$$\mathbf{H} \mathbf{H}^H \phi = \nabla \mathbf{g}_R(\phi) \mu_R - \nabla \mathbf{g}_I(\phi) \mu_I, \quad (28)$$

where $\nabla \mathbf{g}_R(\phi) = 2^{b-1} \text{diag} \left((\phi_1^*)^{2^b-1}, \dots, (\phi_N^*)^{2^b-1} \right)$ and $\nabla \mathbf{g}_I(\phi) = j2^{b-1} \text{diag} \left((\phi_1^*)^{2^b-1}, \dots, (\phi_N^*)^{2^b-1} \right)$. The primal feasibility constraints reduce directly as

$$\frac{\phi_n^{2^b} + (\phi_n^*)^{2^b}}{2} - 1 = \frac{1+1}{2} - 1 = 0, \quad \frac{\phi_n^{2^b} - (\phi_n^*)^{2^b}}{2j} = \frac{1-1}{2j} = 0. \quad (29)$$

For a feasible point such that $\phi_n^{2^b} = 1$, we have $(\phi_n^*)^{2^b-1} = 1/\phi_n^* = \phi_n$. The gradient of the real part of the constraint can be transformed as $\nabla \mathbf{g}_R(\phi)_n = 2^{b-1} \phi_n$. Similarly, for the imaginary part $\nabla \mathbf{g}_I(\phi)_n = j2^{b-1}(\phi_n^*)^{2^b-1} = j2^{b-1} \phi_n$. The stationarity condition becomes

$$\phi_n^* (\mathbf{H} \mathbf{H}^H \phi)_n = 2^{b-1} (\mu_{Rn} + j\mu_{In}), \quad \forall n \in \{1, \dots, N\}. \quad (30)$$

For the discrete phase algorithm, the update rule is given by $\phi_n^{(k+1)} = \exp(j2\pi/2^{b^{(k)}} \left[\vartheta_n^{(k)} 2^{b^{(k)}} / 2\pi + \frac{1}{2} \right])$ where $\vartheta_n^{(k)} = \arg((\mathbf{H} \mathbf{H}^H \phi^{(k)})_n), \forall n \in \{1, \dots, N\}$. As $k \rightarrow \infty$, we

have $b^{(k)} \rightarrow b$, and the limit point satisfies the stationarity conditions associated with the feasible resolution b as

$$\phi_n^* = \exp j \frac{2\pi}{2^b} \left[\frac{\vartheta_n^* 2^b}{2\pi} + \frac{1}{2} \right], \quad \vartheta_n^* = \arg((\mathbf{H}\mathbf{H}^H \phi^*)_n). \quad (31)$$

Let $M_n^* = (\mathbf{H}\mathbf{H}^H \phi^*)_n = |(\mathbf{H}\mathbf{H}^H \phi^*)_n| \exp j \vartheta_n^*$ and $m_n^* = \lfloor \vartheta_n^* 2^b / (2\pi) + \frac{1}{2} \rfloor$, then

$$\phi_n^{* *} M_n^* = |M_n^*| \exp j(\vartheta_n^* - \frac{2\pi}{2^b} m_n^*). \quad (32)$$

We have then

$$\mu_{R_n}^* = \frac{2|M_n^*| \cos(\vartheta_n^* - 2\pi m_n^* / 2^b)}{2^b}, \quad \mu_{I_n}^* = \frac{2|M_n^*| \sin(\vartheta_n^* - 2\pi m_n^* / 2^b)}{2^b}. \quad (33)$$

This ensures the variable resolution schedule algorithm converges to a stationary point associated with the feasible bit resolution.

IV. PRACTICAL PHASE-DEPENDENT AMPLITUDE ELEMENTS

In this section, we extend our work to a more practical model of the phase shifts, where the phase and amplitude are coupled. We derive the projection step and KKT conditions, and we simplify the projection using the structure of the amplitude function.

A. GPM for practical phase-amplitude optimization

In practice, the phase shift vectors are neither continuous nor perfectly unit modulus in amplitude. We employ the model described in [30] that relates the amplitude to the phase by

$$A(\varphi) = (1 - c_1) \left(\frac{\sin(\varphi - c_2) + 1}{2} \right)^{c_3} + c_1, \quad (34)$$

where $\varphi \in \mathbb{R}$ is the phase of the reflecting element, $0 < c_1 < 1$ is the minimum amplitude fraction, $0 < c_2 < \frac{\pi}{2}$ is a constant shift angle and $1 < c_3$ is the constant exponent. Define $\mathcal{A}^N = \{\phi \in \mathbb{C}^N \mid \phi_n = A(\varphi_n) e^{j\varphi_n}, \forall n \in \{1, \dots, N\}\}$ as the practical phase-amplitude set. As discussed, the key step in applying GPM is to specialize the projection to the feasible set. The main optimization step is given as

$$\phi^{(k+1)} = \text{Proj}_{\mathcal{A}^N}(\mathbf{H}\mathbf{H}^H \phi^{(k)}). \quad (35)$$

Since we cannot give the projection directly, we derive it as follows in Lemma 3.

Lemma 3. Let $\mathbf{x} = \mathbf{H}\mathbf{H}^H \phi^{(k)}$ and $\mathbf{y} \in \mathcal{A}^N$. Maximizing the dot product $\text{Re}(\mathbf{x}^H \mathbf{y})$ is equivalent to maximizing

$$\varphi_n = \text{Arg max}_{\varphi} C(\varphi, \vartheta_n) = \text{Arg max}_{\varphi} A(\varphi) \cos(\varphi - \vartheta_n),$$

where $\vartheta_n = \arg(x_n)$ and $y_n = A(\varphi_n) e^{j\varphi_n}$. Hence, the projected phase φ_n satisfies

$$\tan(\varphi_n - \vartheta_n) = \frac{A'(\varphi_n)}{A(\varphi_n)}, \quad \forall n \in \{1, \dots, N\}.$$

Proof. Consider $\mathbf{x} \in \mathbb{C}^N$ and $\mathbf{y} \in \mathcal{A}^N$. To maximize the dot product $\text{Re}(\mathbf{x}^H \mathbf{y})$, we write each term using $x_n = |x_n| e^{j \arg(x_n)}$ and $y_n = A(\varphi_n) e^{j\varphi_n}$, we then obtain

$$\text{Re}(\mathbf{x}^H \mathbf{y}) = \sum_{n=1}^N |x_n| A(\varphi_n) \cos(\varphi_n - \arg(x_n)).$$

Thus, we want to find $\varphi_n = \text{Arg max}_{\varphi} C(\varphi, \vartheta_n) = \text{Arg max}_{\varphi} A(\varphi) \cos(\varphi - \vartheta_n)$, $\forall n \in \{1, \dots, N\}$, where $\vartheta_n = \arg(x_n)$. To determine the projected phase φ_n , we differentiate the function

$$\frac{\partial C}{\partial \varphi_n} = A'(\varphi_n) \cos(\varphi_n - \vartheta_n) - A(\varphi_n) \sin(\varphi_n - \vartheta_n) = 0.$$

Rearranging this expression, we obtain

$$\frac{A'(\varphi_n)}{A(\varphi_n)} = \tan(\varphi_n - \vartheta_n).$$

□

The main step is to maximize the cross term $C(\varphi, \vartheta)$ for a given ϑ . By exploiting the structure of $A(\varphi)$, we aim to obtain a better search region in terms of ϑ , thereby reducing the complexity. It is clear that $A(\varphi)$ is positive and attains its maximum at $\varphi = c_2 + \frac{\pi}{2} + 2m\pi$, $m \in \mathbb{Z}$. The derivative is given by

$$A'(\varphi) = \frac{(c_1 - 1)c_3}{2} \left(\frac{\sin(\varphi - c_2) + 1}{2} \right)^{c_3 - 1} \cos(\varphi - c_2). \quad (36)$$

Similarly, the sign of $A'(\varphi)$ is determined by the sign of $\cos(\varphi - c_2)$. When $\cos(\varphi - \vartheta)$ is positive, we determine a good initial guess for the projected phase based on the interval $[\vartheta - \frac{\pi}{2}, \vartheta + \frac{\pi}{2}]$. We further narrow the search region $\mathbb{I}(\vartheta)$, and define it piecewise as

$$\mathbb{I}(\vartheta) = \begin{cases} [\vartheta - \frac{\pi}{2}, \vartheta], & \vartheta \in (-\pi, c_2 - \frac{\pi}{2}], \\ [\vartheta, \vartheta + \frac{\pi}{2}], & \vartheta \in (c_2 - \frac{\pi}{2}, c_2 + \frac{\pi}{2}], \\ [\vartheta - \frac{\pi}{2}, \vartheta], & \vartheta \in (c_2 + \frac{\pi}{2}, \pi], \end{cases} \quad (37)$$

A justification of this choice can be given by showing that the function is strictly monotonic on one half-interval and has at least one critical point in the other half.

B. Practical amplitude-phase continuous stationarity condition

Since the phase and amplitude are coupled, we can express the constraint in terms of the amplitude of each element as a function of the argument. The Lagrangian for the amplitude-phase constrained problem is then given as

$$\mathcal{L}(\phi, \boldsymbol{\mu}) = \phi^H \mathbf{H}\mathbf{H}^H \phi - \sum_{n=1}^N \mu_n (|\phi_n| - A(\arg(\phi_n))), \quad (38)$$

where $\boldsymbol{\mu} \in \mathbb{R}^N$ is the Lagrange multiplier vector. Using Wirtinger's derivatives

$$\frac{\partial |\phi_n|}{\partial \phi_n^*} = \frac{\phi_n}{2|\phi_n|}, \quad \frac{\partial \arg(\phi_n)}{\partial \phi_n^*} = \frac{j\phi_n}{2|\phi_n|^2}, \quad (39)$$

the gradient of the Lagrangian is

$$\frac{\partial \mathcal{L}}{\partial \phi_n^*} = (\mathbf{H}\mathbf{H}^H \phi)_n - \mu_n \left(\frac{\phi_n}{2|\phi_n|} - \frac{\partial A}{\partial \varphi_n} \frac{j\phi_n}{2|\phi_n|^2} \right), \quad (40)$$

where $\varphi_n = \arg(\phi_n)$, $\forall n \in \{1, \dots, N\}$. The stationarity condition is

$$(\mathbf{H}\mathbf{H}^H \phi)_n = \mu_n \left(\frac{\phi_n}{2|\phi_n|} - \frac{\partial A}{\partial \varphi_n} \frac{j\phi_n}{2|\phi_n|^2} \right). \quad (41)$$

At convergence, we use $\phi_n^* = A(\varphi_n^*)e^{j\varphi_n^*}$ and substitute into the stationarity condition

$$\begin{aligned} (\mathbf{H}\mathbf{H}^H \phi^*)_n &= \mu_n^* \left(\frac{A(\varphi_n^*)e^{j\varphi_n^*}}{2A(\varphi_n^*)} - \frac{jA'(\varphi_n^*)A(\varphi_n^*)e^{j\varphi_n^*}}{2A^2(\varphi_n^*)} \right) \\ &= \frac{\mu_n^* e^{j\varphi_n^*}}{2} \left(1 - j \frac{A'(\varphi_n^*)}{A(\varphi_n^*)} \right). \end{aligned} \quad (42)$$

Let $(\mathbf{H}\mathbf{H}^H \phi^*)_n = |M_n^*|e^{j\vartheta_n^*}$ and divide both sides by $e^{j\vartheta_n^*}$

$$|M_n^*| = \frac{\mu_n^*}{2} e^{j(\varphi_n^* - \vartheta_n^*)} \left(1 - j \frac{A'(\varphi_n^*)}{A(\varphi_n^*)} \right). \quad (43)$$

Let $\Delta\varphi_n^* = \varphi_n^* - \vartheta_n^*$. From the projected phase equation, the optimal phase φ_n^* satisfies $\tan \Delta\varphi_n^* = A'(\varphi_n^*)/A(\varphi_n^*)$. Rewriting the equation for $|M_n^*| = \mu_n^*/2e^{j\Delta\varphi_n^*} (1 - j \tan \Delta\varphi_n^*)$ and using the identity $1 - j \tan \Delta\varphi_n^* = e^{-j\Delta\varphi_n^*}/\cos \Delta\varphi_n^*$, the Lagrange multiplier is

$$\mu_n^* = 2|M_n^*| \cos(\varphi_n^* - \vartheta_n^*). \quad (44)$$

Thus, for the continuous phase-amplitude constraint, we recover a stationarity condition similar to the unit-modulus case, accounting for alignment of the fixed point with its gradient.

V. ERGODIC SNR LOWER BOUNDS

In this section, we utilize the explicit expression of the objective to derive a decomposition of the channel matrix using the distributions of the steering vectors and path gains. By projecting the vector associated with the dominant path onto the unit modulus set, we show that this vector projection is sufficient to achieve the ideal asymptotic SNR scaling of $O(N^2M)$ for all our GPM variants.

A. Channel correlation matrix decomposition

To gain insight into the objective, we write the channel correlation matrix explicitly in terms of the steering vectors as

$$\begin{aligned} \mathbf{H}\mathbf{H}^H &= \frac{NM}{N_G} \sum_{p=1}^{N_G} \sum_{q=1}^{N_G} \alpha_p^* \alpha_q \text{diag}(\mathbf{h}) \mathbf{a}_N^*(\theta_{\text{RIS}_p}) \\ &\quad \times \mathbf{a}_M^T(\theta_{\text{BS}_p}) \mathbf{a}_M^*(\theta_{\text{BS}_q}) \mathbf{a}_N^H(\theta_{\text{RIS}_q}) \text{diag}(\mathbf{h})^H. \end{aligned} \quad (45)$$

Defining the scaled vectors $\tilde{\mathbf{v}}_p = \sqrt{\frac{NM}{N_G}} \text{diag}(\mathbf{h}) \mathbf{a}_N^*(\theta_{\text{RIS}_p}) = \sqrt{\frac{NM}{N_G}} \mathbf{h} \circ \mathbf{a}_N^*(\theta_{\text{RIS}_p})$, the matrix becomes

$$\mathbf{H}\mathbf{H}^H = \sum_{p=1}^{N_G} \sum_{q=1}^{N_G} \alpha_p^* \alpha_q \tilde{\mathbf{v}}_p \mathbf{a}_M^T(\theta_{\text{BS}_p}) \mathbf{a}_M^*(\theta_{\text{BS}_q}) \tilde{\mathbf{v}}_q^H. \quad (46)$$

Define the matrix $\tilde{\mathbf{\Lambda}} \in \mathbb{C}^{N_G \times N_G}$ with entries $\tilde{\Lambda}_{pq} = \alpha_p^* \alpha_q \mathbf{a}_M^T(\theta_{\text{BS}_p}) \mathbf{a}_M^*(\theta_{\text{BS}_q})$, therefore

$$\mathbf{H}\mathbf{H}^H = \tilde{\mathbf{V}} \tilde{\mathbf{\Lambda}} \tilde{\mathbf{V}}^H, \quad (47)$$

where $\tilde{\mathbf{V}} \in \mathbb{C}^{N \times N_G}$ is the matrix of vectors $\tilde{\mathbf{v}}_p$. The diagonal terms satisfy

$$\tilde{\Lambda}_{pp} = \alpha_p^* \alpha_p \mathbf{a}_M^T(\theta_{\text{BS}_p}) \mathbf{a}_M^*(\theta_{\text{BS}_p}) = |\alpha_p|^2 \|\mathbf{a}_M^*(\theta_{\text{BS}_p})\|^2 = |\alpha_p|^2. \quad (48)$$

We recall the expression of the objective, using the decomposition in Eq. (47)

$$\begin{aligned} f(\phi) &= \phi^H \tilde{\mathbf{V}} \tilde{\mathbf{\Lambda}} \tilde{\mathbf{V}}^H \phi \\ &= \sum_{p=1}^{N_G} |\alpha_p|^2 |\tilde{\mathbf{v}}_p^H \phi|^2 + \sum_{p \neq q} \tilde{\Lambda}_{pq} (\tilde{\mathbf{v}}_p^H \phi)^* (\tilde{\mathbf{v}}_q^H \phi). \end{aligned} \quad (49)$$

Let $\phi^{(0)} = e^{j \arg(\tilde{\mathbf{v}}_1)}$, which aligns the RIS phases with the dominant path gain to maximize its energy. For any $p \in \{1, \dots, N_G\}$, we have

$$\begin{aligned} \tilde{\mathbf{v}}_p^H \phi^{(0)} &= \sum_{n=1}^N (\tilde{v}_{pn})^* e^{j \arg(\tilde{v}_{1n})} \\ &= \sqrt{\frac{M}{N_G}} \sum_{n=1}^N |h_n| e^{j(n-1)\omega_{\text{RIS}_p1}}, \end{aligned} \quad (50)$$

where we define the angle difference $\omega_{\text{RIS}_p1} = \pi(\sin(\theta_{\text{RIS}_p}) - \sin(\theta_{\text{RIS}_1}))$. Let us denote $T_p = \sum_{n=1}^N |h_n| e^{j(n-1)\omega_{\text{RIS}_p1}}$. The objective function becomes

$$f(\phi^{(0)}) = \frac{M}{N_G} \left[\sum_{p=1}^{N_G} |\alpha_p|^2 |T_p|^2 + \sum_{\substack{p,q=1 \\ p \neq q}}^{N_G} \tilde{\Lambda}_{pq} T_p^* T_q \right]. \quad (51)$$

Since the path gains are zero mean and uncorrelated ($\mathbb{E}[\alpha_p^* \alpha_q] = 0$ for $p \neq q$), the expected SNR simplifies to the sum of the diagonal terms, which we analyze next. We will show that the first term $|T_1|^2 = \|\mathbf{h}\|_1^2$ dominates the expectation of $|T_p|^2$ for $p \geq 2$.

Lemma 4. Let $\phi^{(0)} = e^{j \arg(\tilde{\mathbf{v}}_1)}$, then, the average objective value satisfies

$$\mathbb{E}[f(\phi^{(0)})] \geq \frac{M}{N_G} \mathbb{E}[|\alpha_1|^2] \mathbb{E}[|T_1|^2].$$

For each $p \geq 2$, the expectation $\mathbb{E}[|T_p|^2]$ is upper bounded as $\mathbb{E}[|T_p|^2] = O(N)$, while the dominant term satisfies

$$\mathbb{E}[|T_1|^2] \geq \left(N + \frac{\pi}{4} N(N-1) \right) \geq \frac{\pi}{4} N^2.$$

Hence, in the large N regime, the ergodic SNR contribution of the first path dominates those of the remaining paths.

The proof of this lemma can be found in Appendix A.

B. Unit modulus continuous lower bound

Due to the non-decreasing property of the iterates, we can choose $\phi^{(0)} = \text{Proj}_{\mathcal{F}_N}(\tilde{\mathbf{v}}_1) = e^{j \arg(\tilde{\mathbf{v}}_1)}$ and apply the lemma's results directly to find a minimum ergodic SNR bound. According to Lemma 4, we have

$$\mathbb{E}[f(\phi^{(0)})] \geq \mathbb{E}[|\alpha_1|^2] \frac{\pi N^2 M}{4 N_G}. \quad (52)$$

The gains are distributed as $\alpha_p \sim \mathcal{CN}(0, 1)$ therefore $|\alpha_p| \sim \text{Exp}(1)$. According to [31], the expectation of the maximum of N_G i.i.d. exponential variables is $H_{N_G} = \sum_{p=1}^{N_G} \frac{1}{p}$. Finally, for any fixed point ϕ^* , we take the conservative lower bound on the expectation as

$$\mathbb{E}[f(\phi^*)] \geq \mathbb{E}[f(\phi^{(0)})] \geq \frac{H_{N_G} \pi N^2 M}{4 N_G}, \quad (53)$$

which guarantees a minimum achievable SNR on average under this initialization.

C. Unit modulus discrete lower bound

Next, we quantify the performance loss when using discrete shifts. We choose $\phi^{(0)} \in \mathcal{F}_b^N = \text{Proj}_{\mathcal{F}_b^N}(\tilde{\mathbf{v}}_1)$. This means $\phi_n^{(0)} = e^{j(\arg(\tilde{v}_{1n}) + E_b(\arg(\tilde{v}_{1n})))}$, where $E_b(\arg(\tilde{v}_{1n}))$ is the quantization phase error per element of $\tilde{\mathbf{v}}_1$. Using $\tilde{v}_{1n} = |\tilde{v}_{1n}|e^{j\arg(\tilde{v}_{1n})}$

$$\tilde{\mathbf{v}}_1^H \phi^{(0)} = \sum_{n=1}^N |\tilde{v}_{1n}| e^{j(\arg(\tilde{v}_{1n}) - \arg(\tilde{v}_{1n}) + E_b(\arg(\tilde{v}_{1n})))}. \quad (54)$$

Lower-bounding the dot product and the quantization error as

$$\begin{aligned} |\tilde{\mathbf{v}}_1^H \phi^{(0)}|^2 &\geq \left(\sum_{n=1}^N |\tilde{v}_{1n}| \cos(E_b(\arg(\tilde{v}_{1n}))) \right)^2 \\ &\geq \cos^2\left(\frac{\pi}{2^b}\right) \|\tilde{\mathbf{v}}_1\|_1^2. \end{aligned} \quad (55)$$

Under this initialization, every fixed point ϕ^* satisfying KKT conditions is lower-bounded as

$$\mathbb{E}[f(\phi^*)] \geq \mathbb{E}[f(\phi^{(0)})] \geq \cos^2\left(\frac{\pi}{2^b}\right) \frac{H_{NG} \pi N^2 M}{4N_G}, \quad (56)$$

which proves that, despite the discrete phase constraint, a notable fraction of the ideal $O(N^2 M)$ scaling SNR can be achieved for $b \geq 2$.

D. Practical amplitude-phase continuous lower bound

Finally, for the practical amplitude-phase model we let $\phi^{(0)} = \text{Proj}_{\mathcal{A}^N}(\tilde{\mathbf{v}}_1)$, such that $\phi_n^{(0)} = A(\varphi_n) e^{j\varphi_n}$. We then have

$$|\tilde{\mathbf{v}}_1^H \phi^{(0)}|^2 = \left| \sum_{n=1}^N A(\varphi_n) |\tilde{v}_{1n}| e^{j(\varphi_n - \arg(\tilde{v}_{1n}))} \right|^2. \quad (57)$$

Using the projected phase condition $\varphi_n = \arg(\tilde{v}_{1n}) + \arctan(A'(\varphi_n)/A(\varphi_n))$, we can write

$$\begin{aligned} |\tilde{\mathbf{v}}_1^H \phi^{(0)}|^2 &\geq \left(\sum_{n=1}^N A(\varphi_n) |\tilde{v}_{1n}| \cos\left(\arctan\left(\frac{A'(\varphi_n)}{A(\varphi_n)}\right)\right) \right)^2 \\ &= \left(\sum_{n=1}^N \frac{A(\varphi_n) |\tilde{v}_{1n}|}{\sqrt{1 + (A'(\varphi_n)/A(\varphi_n))^2}} \right)^2 \\ &\geq \left(\min_{\varphi} R(\varphi) \sum_{n=1}^N |\tilde{v}_{1n}| \right)^2, \end{aligned} \quad (58)$$

where we define the ratio $R(\varphi) = A(\varphi)^2 / \sqrt{A(\varphi)^2 + A'(\varphi)^2}$. This ratio encapsulates the total amplitude loss, combining the physical amplitude $A(\varphi)$ with the loss from phase projection. The function $A(\varphi)$ attains its minimum c_1 when $\sin(\varphi - c_2) = -1$, where the derivative $A'(\varphi)$ vanishes. At these points, $R(\varphi) = c_1^2 / |c_1| = c_1$, which represents its global minimum since $R(\varphi) \leq A(\varphi)$ always. Finally, for any fixed point ϕ^* under this initialization, we have

$$\mathbb{E}[f(\phi^*)] \geq \mathbb{E}[f(\phi^{(0)})] \geq c_1^2 \frac{H_{NG} \pi N^2 M}{4N_G}. \quad (59)$$

This final bound demonstrates that even with phase-dependent amplitude variations, the favorable $O(N^2 M)$ scaling is maintained, scaled by a factor c_1^2 determined by the minimum amplitude reflection.

VI. COMPUTATIONAL COMPLEXITY ANALYSIS

In this section, we analyze the computational complexity and iteration requirements of our proposed methods and compare them with existing continuous and discrete benchmarks, both under unit modulus and amplitude-phase coupling constraints.

A. Unit modulus continuous phase problem

For the continuous case, we compare our algorithm with other gradient methods. Gradient descent/ascent has been applied to the unit modulus constraint using an alternating approach described in [10], [16], where the gradient update is performed and then projected onto the feasible set. The projected gradient ascent (PGA) update can be given as

$$\begin{aligned} \phi^{(k+1)} = \text{Arg max}_{\phi \in \mathcal{Q}} \left\{ f(\phi^{(k)}) + \left\langle \nabla f(\phi^{(k)}), \phi - \phi^{(k)} \right\rangle \right. \\ \left. - \frac{1}{2\eta} \|\phi - \phi^{(k)}\|^2 \right\}. \end{aligned} \quad (60)$$

Similar to Eq.(11), the update simplifies to $\phi^{(k+1)} = \text{Arg max}_{\phi \in \mathcal{Q}} \left\langle \nabla f(\phi^{(k)}) + \frac{1}{\eta} \phi^{(k)}, \phi \right\rangle$. We recover the conventional update as $\phi^{(k+1)} = \text{Proj}_{\mathcal{Q}}(\phi^{(k)} + \eta \nabla f(\phi^{(k)}))$. When $\mathcal{Q} = \mathcal{F}_{\infty}^N$ the PGA update for (9) is given by

$$\phi^{(k+1)} = \exp j \arg \left(\phi^{(k)} + \eta \mathbf{H} \mathbf{H}^H \phi^{(k)} \right), \quad (61)$$

And when step size $\eta \gg 1$, the update simplifies to $\phi^{(k+1)} = \exp j \arg(\mathbf{H} \mathbf{H}^H \phi^{(k)})$, which is equivalent to the GPM update. The PGA update incorporates a regularization term of $1/\eta$ that stabilizes the gradient. For a step-size η satisfying $\eta \leq 1/L_f$ where L_f is the Lipschitz constant for f , we upper bound the required number of iterations K such that the distance between the iterates is less than some tolerance ϵ using the result found in [16, Appendix B]

$$K_{\text{PGA}} \geq \frac{2(f(\phi^*) - f(\phi^{(0)}))}{\eta \epsilon}. \quad (62)$$

While the convergence guarantee holds for step-sizes $\eta \leq 1/L_f$, empirically, [10] shows that the distance between the iterates converges much faster, indicating that the smoothness constant along the optimization trajectory is much smaller than L_f . Taking $\eta \rightarrow \infty$ effectively removes the memory of the previous iterate and forces alignment with the steepest ascent direction in the subspace spanned by $\mathbf{H} \mathbf{H}^H$. In the case of a rank-deficient matrix, this behavior is beneficial as the iterates are driven toward the dominant eigenspace, and empirical convergence occurs in fewer iterations. The p-norm radiant search (p-GS) method [32] is a modified gradient ascent algorithm that utilizes a gradient term with an increasing p -norm with a step dp . The optimal direction is obtained using line search over Θ angles. Although p-GS uses an adaptive

p -norm for early exploration and stabilizes over iterations, it adds computational overhead due to multiple candidates at each iteration. Since the gradient is computed with complexity $O(MN)$, the overall complexity for PGA is $O(KNM)$, and $O(K(\Theta NM + NP))$ for p-GS, where Θ denotes the number of search angles and P is the number of increases of p detailed in [33].

B. Unit modulus discrete phase problem

For the discrete case, we compare exact optimization and different gradient methods. The MaRLI-I Algorithm relaxes the update described in [19, Eq. (74)] as

$$\phi_n^{(k+1)} = \left| (\mathbf{H}\mathbf{H}^H \phi^{(k)})_n \right| e^{-\nu_1 k} e^{j \frac{2\pi}{2^b} (\lfloor m_n^{(k)} \rfloor + \{m_n^{(k)}\} e^{-\nu_2 k})}. \quad (63)$$

$(\mathbf{H}\mathbf{H}^H \phi^{(k)})_n$ is the n th element of the gradient and where $m_n^{(k)} = 2^b \arg((\mathbf{H}\mathbf{H}^H \phi^{(k)})_n) / 2\pi + \frac{1}{2}$ represents the quantized phase element before rounding and $\{\cdot\}$ denotes the fractional part. The constants $\nu_1, \nu_2 > 1$ control the rate at which the update is refined from an unconstrained setting to the discrete phase set. To ensure that the relaxation factors $e^{-\nu_1 k}$ and $e^{-\nu_2 k}$ decay below 10^{-6} , the number of iterations $K_{\text{MaRLI-I}}$ must satisfy

$$K_{\text{MaRLI-I}} \geq \max \left\{ \frac{-\ln(10^{-6})}{\nu_1}, \frac{-\ln(10^{-6})}{\nu_2} \right\}. \quad (64)$$

Our variable resolution GPM-Disc projects onto the set $\mathcal{F}_{b^{(k)}}^N$, where $b^{(k)}$ starts from B and approaches b . This requires a minimum number of iterations to reach the feasible resolution and then converge to the stationary point. Thus, we have

$$K_{\text{GPM-Disc}} \geq \left\lceil \frac{\ln(B - b)}{\gamma} \right\rceil, \quad (65)$$

which represents a trade-off between the decay rate γ and preventing early convergence to a non-efficient stationary point. Both methods have a complexity of $O(KNM)$, where the number of iterations depends on the decay parameters.

C. Phase-amplitude problem

a) Continuous phase optimization via golden section search: For the phase-amplitude problem, a simplified optimized phase (SOP) algorithm was introduced in [9, Alg. 2], where the phase-coupled quadratic optimization is addressed via iterative element-wise updates. Each phase is determined by scalar maximization. However, the authors do not specify the method used in the continuous case, nor analyze the structure of the objective. In contrast, our GPM algorithm updates by solving a scalar maximization problem with a structured form $C(\varphi, \vartheta_n)$, where $\vartheta_n = \arg((\mathbf{H}\mathbf{H}^H \phi^{(k)})_n)$. This function is guaranteed to have a maximum within $\mathbb{I}(\vartheta_n)$, allowing us to solve the subproblem using golden section search. The golden section method iteratively narrows the interval of length Δ by a constant factor $r = (\sqrt{5} - 1)/2$. After G iterations, the search interval is reduced to Δr^G , and to ensure accuracy within tolerance ϵ , we require

$$G_\Delta = \left\lceil \frac{\log(\epsilon/\Delta)}{\log(r)} \right\rceil, \quad (66)$$

where $\Delta = \frac{\pi}{2}$. Each projection costs $O(G_{\pi/2})$ per element, leading to an overall projection complexity of $O(NG_{\pi/2})$ per iteration. For comparison, we utilize the same method for the SOP algorithm, assuming the amplitude function is periodic over $[0, 2\pi)$. The resulting complexity is then $O(K(N^2 + NG_{2\pi}))$ for SOP and $O(K(NM + NG_{\pi/2}))$ for our GPM-Prac-Cont.

b) Discrete phase optimization via exhaustive search:

In the discrete setting, the phase φ is restricted to the set $\varphi \in \mathcal{P}_b = \{ \frac{2\pi m}{2^b} \mid m = 0, 1, \dots, 2^b - 1 \}$. Since the candidate values are finite and known in advance, we precompute $A(\mathcal{P}_b)$ once, avoiding redundant evaluations during the search. This allows us to extract the relevant precomputed values directly, further optimizing the search. As shown for the continuous case, the derivative changes sign within the half interval $\mathbb{I}(\vartheta)$ and is strictly monotonic within the other half. Thus, we need to check all the values within $\mathbb{I}(\vartheta)$ and the boundary value in the other half. Because $\mathcal{P}_b$ consists of 2^b values that span $[0, 2\pi)$, the number of candidates within the quantized interval $\mathbb{I}(\vartheta)$ of length $\Delta = \frac{\pi}{2}$ is at most $\frac{2^b \pi/2}{\Delta} + 1 = 2^{b-2} + 1$, leading to a projection cost of $O(N2^{b-2} + 2)$ and an overall complexity for GPM-Prac-Disc of $O(K(NM + N2^{b-2}))$. The corresponding complexities for all constraints are summarized in Table I.

TABLE I: Complexity comparison of continuous and discrete methods

Continuous methods		
Method	Complexity	Description
PGA [10]	$O(KNM)$	Step-size η
p-GS [32]	$O(K(\Theta NM + NP))$	Θ search angles, p -norm
GPM-Cont	$O(KNM)$	Continuous GPM
SOP [9, Alg. 2]	$O(K(N^2 + NG_{2\pi}))$	Simplified optimal phase
GPM-Prac-Disc	$O(K(NM + NG_{\pi/2}))$	Golden section search
Discrete methods		
Method	Complexity	Description
RDQF [4]	$O((N2^{b-1})^2 N_G)$	Rank-deficient quadratic
DCM [8]	$O(NM + N2^b)$	Dominant column matching
MaRLI-I [19]	$O(KNM)$	Relaxed power method like
GPM-Disc	$O(KNM)$	Variable resolution GPM
GPM-Prac-Cont	$O(K(NM + N2^{b-2}))$	Reduced discrete search

VII. NUMERICAL RESULTS

In this section, we present numerical convergence results across channel configurations, comparing initialization, run-time, and iteration count. Finally, we validate the theoretical lower bound on the ergodic SNR achieved by our algorithms.

We evaluate the convergence behavior of three continuous-phase unit modulus optimization algorithms, PGA, p-GS, and our proposed GPM-Cont, under identical random initializations $\phi^{(0)}$ in $\mathcal{F}_\infty^N$. For each realization, the angles of arrival and departure at the RIS and BS are drawn from $\mathcal{U}(\theta^{\text{LoS}} - \delta, \theta^{\text{LoS}} + \delta)$, where $\theta_{\text{RIS}}^{\text{LoS}} = \theta_{\text{BS}}^{\text{LoS}} = \frac{\pi}{3}$, $\theta_{\text{UE}}^{\text{LoS}} = -\frac{\pi}{3}$ and $\delta = \frac{\pi}{18}$. The path gains α_p and β_i are sampled from $\mathcal{CN}(0, 1)$. We fix the number of propagation paths to $N_G = 3$ for the BS-RIS link and $N_h = 4$ for the RIS-UE link in

all experiments. The Lipschitz constant of the objective is given as the largest eigenvalue $L_f = \lambda_1$, computed offline, and PGA is run with step sizes $\eta = 1/L_f$, $\eta = 5/L_f$, and $\eta = 10/L_f$. For GPM-Prac-Cont we draw random phases $\varphi \in (-\pi, \pi]^N$ and then take $\phi_n^{(0)} = A(\varphi_n)e^{j\varphi_n}$, where $(c_1, c_2, c_3) = (0.2, 0.43\pi, 1.6)$ are the constants for the phase amplitude function. The maximum number of search iterations for the golden search method is $G_{\pi/2} = 20$ and $G_{2\pi} = 27$.

Fig. 1 compares the convergence for the continuous methods for different channel configurations. The objective value versus iteration shows that PGA, p-GS, and GPM-Cont converge to nearly the same value. We note that GPM-Cont approaches the converged point the fastest within a few iterations, while PGA reveals that $\eta = 1/L_f$ requires significantly more iterations than $\eta = 5/L_f$ and $\eta = 10/L_f$. pGS only falls slightly behind on convergence speed and converges within a few added iterations. This trend persists across all configurations, including the large-dimensional setting in Fig. 1c, highlighting the scalability of our approach. Notably, the performance gap between GPM-Cont and the slower PGA with small η widens as the dimension grows, indicating that aggressive updates (as in GPM-Cont or PGA with large η) are crucial for high-dimensional settings. For GPM-Prac-Disc, it maintains the non-decreasing improvement using Golden Section Search. SOP-Prac-Cont maintains the same performance under the same initialization, settling at a notable fraction of the ideal unit modulus case.

Fig. 2 illustrates the convergence behavior of the discrete methods. We compare MaRLI-I and DCM-Disc with our variable-resolution GPM-Disc, using $B = 15$ and $\gamma = 1$. The early values of MaRLI-I are cropped because its relaxed update temporarily violates the unit-modulus constraint. We also include the exhaustive search (RDQF) in the first subplot only, as its complexity grows prohibitively for larger dimensions. All discrete iterative methods share the same random initialization $\phi^{(0)} \in \mathcal{F}_b^N$, using a bit resolution $b = 2$ and the same channel realizations. For the discrete practical phase-amplitude problem, we compare our GPM-Prac-Disc with DCM-Prac.

As shown in Fig. 2a, RDQF provides the globally optimal solution for small-scale systems. In the same plot, MaRLI-I, GPM-Disc, and DCM-Disc all converge to the same final objective value, at different rates. For larger configurations shown in Fig. 2b and Fig. 2c, MaRLI-I and GPM-Disc again converge to the same point. However, MaRLI-I takes significantly more iterations to return to the discrete unit modulus set and settle at a point. In contrast, our GPM-Disc, with its variable-resolution schedule, enables smoother and less drastic updates. This leads to faster convergence while maintaining robustness in high dimensions. In the practical phase-amplitude case, our proposed GPM-Prac-Disc consistently outperforms DCM-Prac across all scenarios. It maintains the desirable behavior of GPM and achieves a significant fraction of the objective value attained by the unit modulus algorithms, despite the amplitude loss. To quantify the computational efficiency, we first execute all iterative algorithms with a maximum of $K_{\max} = 40$ iterations. For each method, we then determine the minimum number of iterations K_{req} required to satisfy the relative

improvement criterion $|f(\phi^{(k+1)}) - f(\phi^{(k)})|/f(\phi^{(k)}) < \epsilon$, where $\epsilon = 10^{-4}$. Once K_{req} is determined under a fixed random seed, we rerun each iterative algorithm and measure the wall-clock time of the main update loop. Non-iterative methods (RDQF and DCM variants) are timed in a single execution under the same seed. Table II summarizes the required iterations and execution times over different (M, N) configurations. From Table II, we observe that GPM-Cont converges in only 2-3 iterations and completes in under 2.5ms even for the largest dimensions, thanks to its direct update rule. p-GS needs one extra iteration on average, but takes up to 53.8ms because of its angle search overhead. PGA can require up to 40 iterations and around 22.4ms total using a step size $\eta = 1/L_f$. GPM-Prac-Cont, which handles phase amplitude coupling via golden-section search, still finishes in under 60.3 ms and closely matches GPM-Cont's iteration count for the larger dimensions, demonstrating that practical constraints add little overhead. Additionally, it outperforms SOP-Prac-Cont in terms of convergence speed as it takes more than double the time to converge, even with the same number of iterations. In the discrete case, GPM-Disc finds high-quality solutions in under 5 iterations and around 4.2ms, while MaRLI-I uses, on average, 12 iterations and up to 12.7ms to enforce feasibility. RDQF is exact but takes several seconds, making it impractical beyond the smallest dimensions. Greedy DCM-Disc and DCM-Prac-Disc run almost instantly (sub-7ms) but settle at lower objectives. Our GPM-Prac-Disc balances speed and solution quality, converging in fewer than 5 iterations and under 46.5ms, proving its feasibility for hardware-constrained deployments. We verify asymptotic performance by plotting the expected limit point for each GPM variant versus N^2M , reflecting ideal scaling. For each channel of the 10^3 channel realization, we initialize by projecting the vector $\tilde{\mathbf{v}}_1$ associated with the dominant path gain $|\alpha_1|$ of the channel correlation matrix onto the feasible set. The solid markers correspond to the average of limit points of GPM-Cont, GPM-Disc, GPM-Prac-Cont, and GPM-Prac-Disc. The dashed curves represent the theoretical lower bounds. For GPM-Disc, the bound is scaled by $\cos^2(\pi/2^b)$ (with bit resolution $b = 2$), and for GPM-Prac-Cont, the bound is multiplied by c_1^2 to account for amplitude loss, where $c_1 = 0.2$. We evaluate the following configurations $(M, N) = [(2, 8), (4, 16), (16, 64), (32, 128), (64, 512), (128, 1024)]$. Fig. 3 shows that GPM-Cont almost exactly recovers the ideal $O(N^2M)$ SNR scaling. With 2-bit resolution, GPM-Disc retains 81.82% of the continuous SNR, well above the $\cos^2(\pi/4) = 50\%$ bound. GPM-Prac-Cont exceeds its $c_1^2 = 4\%$ guarantee by achieving 39.11%, while GPM-Prac-Disc reaches 26.97% of the continuous unit-modulus SNR, demonstrating viable performance under the most realistic RIS constraints.

VIII. CONCLUSIONS

We proposed a unified framework for RIS-assisted wireless systems under increasingly realistic constraints. For the continuous unit modulus problem, we showed that the GPM

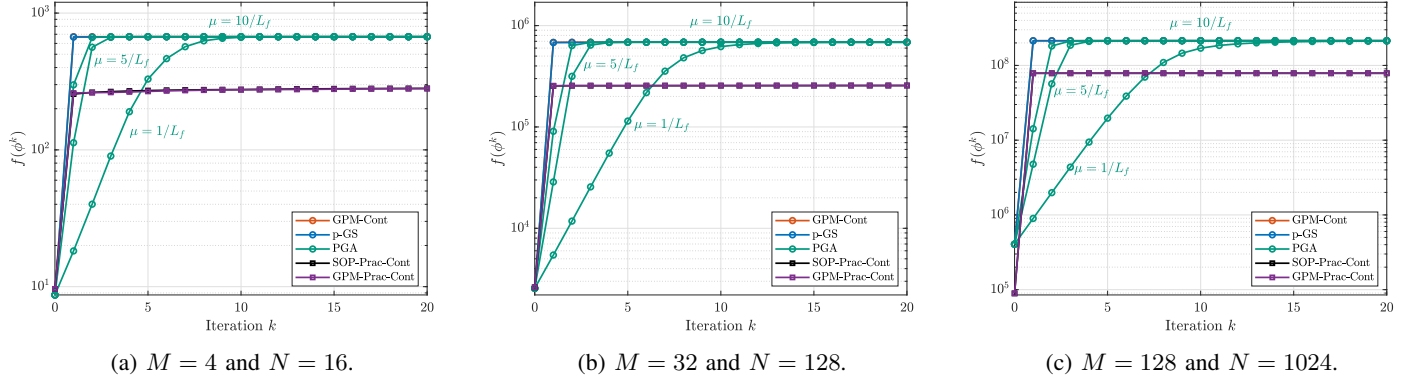


Fig. 1: Iteration comparison of PGA with $\eta = 1/L_f$, $5/L_f$, $10/L_f$, p-GS, our proposed GPM-Cont, and GPM-Prac-Cont for different channel configurations.

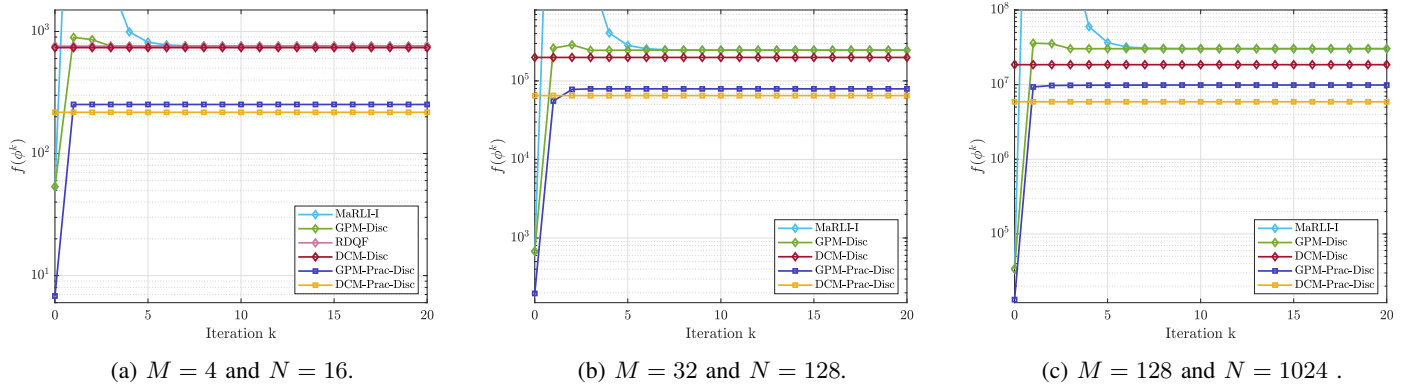


Fig. 2: Iteration comparison of MaRLI-I, GPM-Disc with $B = 15$, $\gamma = 1$, RDQF, and DCM-Disc as well as GPM-Disc-Prac and DCM-Disc-Prac, for different channel configurations, with $b = 2$.

TABLE II: Performance of optimization methods over different channel configurations

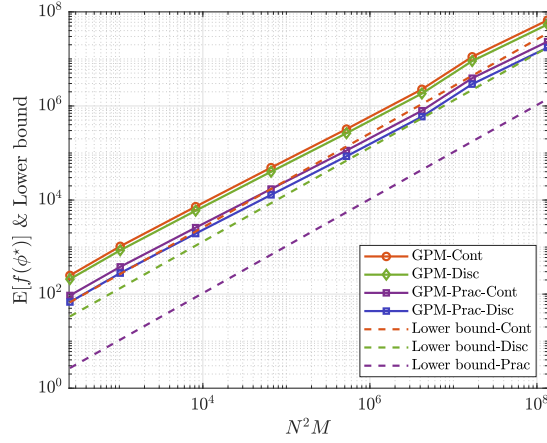
Method	(M, N)		$(2, 8)$		$(4, 16)$		$(16, 64)$		$(32, 128)$		$(64, 512)$		$(128, 1024)$	
			Iters	Time	Iters	Time	Iters	Time	Iters	Time	Iters	Time	Iters	Time
GPM-Cont			2	0.6ms	2	0.7ms	3	0.7ms	3	0.7ms	3	1.3ms	3	2.5ms
p-GS [32]			3	13.2ms	3	14.6ms	3	15.4ms	4	20.9ms	3	26.5ms	3	53.8ms
PGA [10]			14	2.4ms	14	3.0ms	17	2.7ms	21	3.2ms	40	9.2ms	36	22.4ms
SOP-Prac-Cont [9]			10	11.9ms	29	38.4ms	3	16.0ms	14	92.0ms	3	74.9ms	3	147.9ms
GPM-Prac-Cont			27	8.4ms	40	24.0ms	3	7.0ms	3	10.5ms	3	30.1ms	3	60.3ms
MaRLI-I [19]			11	2.2ms	12	2.7ms	12	2.7ms	12	3.0ms	12	6.0ms	13	12.7ms
RDQF [4]			—	349ms	—	324.3s	—	—	—	—	—	—	—	—
GPM-Disc			4	1.7ms	4	1.9ms	5	3.1ms	4	1.7ms	5	4.0ms	4	4.2ms
DCM-Disc [8]			—	0.4ms	—	0.4ms	—	0.5ms	—	0.9ms	—	2.5ms	—	5.2ms
GPM-Prac-Disc			4	2.8ms	2	2.9ms	4	6.0ms	3	8.2ms	3	25.3ms	3	46.5ms
DCM-Prac-Disc [8]			—	0.3ms	—	0.3ms	—	0.6ms	—	1.0ms	—	3.2ms	—	6.9ms

update is the limit of the gradient ascent update as the step size goes to infinity. Through empirical analysis, we showed that this characteristic, driven by the objective function's low-rank structure and the resulting sharper alignment with the gradient direction, enables our GPM update to converge the fastest to the fixed point, verifying stationarity conditions.

Extending this, we explicitly derived the projection steps and analyzed KKT conditions for continuous, discrete, and practical phase-amplitude coupled RIS elements. For the discrete unit modulus problem, we introduced a variable bit

resolution schedule that starts with a near-continuous phase and transitions into the feasible set, allowing for smoother convergence and avoiding early non-efficient stationary points. For the practical phase-amplitude model, we presented an efficient projection step that leverages the structure of the amplitude function, allowing for element-wise optimization via golden section search for continuous phases and reduced exhaustive search for the discrete setting.

Our analysis and simulations show that GPM-Cont is faster and more computationally efficient than other gradient meth-


 Fig. 3: Ergodic limit point and lower bounds versus $N^2 M$.

ods. For discrete phase shifts, GPM-Disc with its variable-resolution approach converges much quicker than existing relaxed methods while maintaining feasibility. Our practical algorithms, GPM-Prac-Cont and GPM-Prac-Disc, showcase viability under realistic amplitude-phase coupling, balancing high solution quality with efficient computation. We theoretically derived and empirically confirmed that all GPM variants achieve the expected $O(N^2 M)$ ergodic SNR scaling by aligning with the dominant path gain, quantifying real-world limitations like discretization and amplitude losses.

This work thus provides a versatile, computationally efficient, and theoretically sound set of tools for maximizing RIS performance under diverse constraints. As future work, our framework can be extended to study other performance metrics, such as SINR in multi-user interference-limited scenarios and sum rate maximization, which are crucial for practical multi-user communication systems.

APPENDIX A

PROOF OF DOMINANT PATH EXPECTATION LEMMA

Assumption 1. The path gains $\{\alpha_p\}_{p=1}^{N_G} \sim \mathcal{CN}(0, 1)$ and $\{\beta_i\}_{i=1}^{N_h} \sim \mathcal{CN}(0, 1)$ are i.i.d. complex Gaussian. Without loss of generality, we assume the gains α_p are sorted in descending order of magnitude, i.e., $|\alpha_1|^2 \geq |\alpha_2|^2 \geq \dots \geq |\alpha_{N_G}|^2$.

Assumption 2. The steering angles at the BS, RIS, and UE are all i.i.d. and follow a uniform distribution centered around their respective LoS angles. Specifically, for some deviation $\delta > 0$ satisfying $0 < |\theta^{\text{LoS}}| + \delta < \frac{\pi}{2}$, we have

$$\theta \sim \mathcal{U}(\theta^{\text{LoS}} - \delta, \theta^{\text{LoS}} + \delta),$$

where θ denotes any of $\{\theta_{BSp}\}_{p=1}^{N_G}$, $\{\theta_{RISp}\}_{p=1}^{N_G}$, or $\{\theta_{UEi}\}_{i=1}^{N_h}$.

Proof. For each n , we have that $h_n = \sum_{i=1}^{N_h} \beta_i e^{j(n-1)\pi \sin(\theta_{UEi})} / \sqrt{N_h}$, under Assumptions 1 and 2. Conditioned on $\theta_{UE} = \{\theta_{UEi}\}_{i=1}^{N_h}$, we have $|h_n| \mid \theta_{UE} \sim \text{Rayleigh}(\sqrt{1/2})$ and $|h_n|^2 \mid \theta_{UE} \sim \text{Exp}(1)$. Define $\|\mathbf{h}\|_1^2 = (\sum_{n=1}^N |h_n|)^2 = |T_1|^2$. Conditioned on θ_{UE} ,

the $|h_n|$ are identically distributed Rayleigh variables but not independent. Thus we have

$$\mathbb{E}[\|\mathbf{h}\|_1^2 \mid \theta_{UE}] = \sum_{n=1}^N \mathbb{E}[|h_n|^2 \mid \theta_{UE}] + \sum_{n \neq m} \mathbb{E}[|h_n| |h_m| \mid \theta_{UE}].$$

The correlation coefficient between h_n and h_m is $\mathbb{E}[h_n h_m^* \mid \theta_{UE}] = \rho_{nm}$, where $\rho_{nm} = \sum_{i=1}^{N_h} e^{j(n-m)\pi \sin(\theta_{UEi})} / N_h$. For two correlated, zero-mean complex Gaussian variables, the expected product of their magnitudes can be expressed using the hypergeometric function [34]

$$\mathbb{E}[|h_n| |h_m| \mid \theta_{UE}] = \frac{\pi}{4} {}_2F_1\left(-\frac{1}{2}, -\frac{1}{2}; 1; |\rho_{nm}|^2\right).$$

We recall the expression of the hypergeometric function

$${}_2F_1\left(-\frac{1}{2}, -\frac{1}{2}; 1; x\right) = \sum_{k=0}^{\infty} \frac{(-\frac{1}{2})_k (-\frac{1}{2})_k}{(1)_k} \frac{x^k}{k!}.$$

Using the identity $(a)_k = \frac{\Gamma(a+k)}{\Gamma(a)}$, we obtain

$${}_2F_1\left(-\frac{1}{2}, -\frac{1}{2}; 1; x\right) = \sum_{k=0}^{\infty} \frac{\Gamma(k - \frac{1}{2})^2}{\Gamma(-\frac{1}{2})^2} \frac{x^k}{(k!)^2}.$$

Since ${}_2F_1(-\frac{1}{2}, -\frac{1}{2}; 1; x) \geq 1$ for $0 \leq x \leq 1$, we obtain the lower bound $\mathbb{E}[|h_n| |h_m| \mid \theta_{UE}] \geq \frac{\pi}{4}$. Taking the expectation over θ_{UE} and summing over all $N(N-1)$ cross terms yields

$$\mathbb{E}[\|\mathbf{h}\|_1^2] = \mathbb{E}[\mathbb{E}[\|\mathbf{h}\|_1^2 \mid \theta_{UE}]] \geq N + \frac{\pi}{4} N(N-1).$$

For $p \geq 2$, recall $T_p = \sum_{n=1}^N |h_n| e^{j(n-1)\omega_{\text{RIS}p1}}$. Its expected squared magnitude is

$$\mathbb{E}[|T_p|^2] = \sum_{n=1}^N \mathbb{E}[|h_n|^2] + \sum_{n \neq m} \mathbb{E}[|h_n| |h_m|] \mathbb{E}[e^{j(n-m)\omega_{\text{RIS}p1}}].$$

Define the integral $I_n = \frac{1}{2\delta} \int_{\theta^{\text{LoS}} - \delta}^{\theta^{\text{LoS}} + \delta} e^{jn\pi \sin(\theta)} d\theta$, with $I_0 = 1$. Using integration by parts with $u(\theta) = \frac{1}{\cos(\theta)}$ and $dv = \cos(\theta) e^{jn\pi \sin(\theta)} d\theta$, we compute

$$du = \frac{\sin(\theta)}{\cos^2(\theta)} d\theta, \quad v = \frac{e^{jn\pi \sin(\theta)}}{jn\pi}.$$

Then,

$$I_n = \frac{1}{2\delta jn\pi} \left[\frac{e^{jn\pi \sin(\theta)}}{\cos(\theta)} \right]_{\theta^{\text{LoS}} - \delta}^{\theta^{\text{LoS}} + \delta} - \frac{1}{2\delta jn\pi} \int_{\theta^{\text{LoS}} - \delta}^{\theta^{\text{LoS}} + \delta} \frac{e^{jn\pi \sin(\theta)} \sin(\theta)}{\cos^2(\theta)} d\theta.$$

By Assumption 2, we have that $|\cos(\theta)| \geq \cos(|\theta^{\text{LoS}}| + \delta) > 0$ on the interval and $|e^{jn\pi \sin(\theta)}| = 1$, we bound

$$\begin{aligned} |I_n| &\leq \frac{1}{2\delta n\pi} \left(\frac{2}{\cos(|\theta^{\text{LoS}}| + \delta)} + \frac{2\delta}{\cos(|\theta^{\text{LoS}}| + \delta)^2} \right) \\ &= \frac{1}{n\pi} \left(\frac{1}{\delta \cos(|\theta^{\text{LoS}}| + \delta)} + \frac{1}{\cos(|\theta^{\text{LoS}}| + \delta)^2} \right). \end{aligned}$$

Using $\mathbb{E}[|h_n|^2] = 1$, $\mathbb{E}[|h_n| |h_m|] \leq 1$, and $\mathbb{E}[e^{jn\omega_{p1}}] = |I_n|^2$, we obtain

$$\mathbb{E}[|T_p|^2] \leq N + 2 \sum_{n=1}^{N-1} (N-n) |I_n|^2.$$

Let $\tau = \frac{1}{\delta \cos(|\theta \text{LoS}| + \delta)} + \frac{1}{\cos(|\theta \text{LoS}| + \delta)^2}$, we have then $|I_n| \leq \frac{\tau}{n\pi}$, which leads to

$$2 \sum_{n=1}^{N-1} (N-n) |I_n|^2 \leq \frac{2N\tau^2}{\pi^2} \sum_{n=1}^{\infty} \frac{1}{n^2} = \frac{N\tau^2}{3}.$$

This gives the final bound $\mathbb{E}[|T_p|^2] \leq N + \frac{N\tau^2}{3} = O(N)$. $\square$

REFERENCES

- [1] E. Basar, M. Di Renzo, J. De Rosny, M. Debbah, M.-S. Alouini, and R. Zhang, "Wireless communications through reconfigurable intelligent surfaces," *IEEE Access*, vol. 7, pp. 116 753–116 773, 2019.
- [2] Q. Wu and R. Zhang, "Beamforming optimization for wireless network aided by intelligent reflecting surface with discrete phase shifts," *IEEE Transactions on Communications*, vol. 68, no. 3, pp. 1838–1851, 2020.
- [3] Y. Zhang, J. Zhang, M. Di Renzo, H. Xiao, and B. Ai, "Performance analysis of ris-aided systems with practical phase shift and amplitude response," *IEEE Transactions on Vehicular Technology*, vol. 70, no. 5, pp. 4501–4511, 2021.
- [4] J. Ye, A. Kammoun, and M.-S. Alouini, "Optimal phase shift solution and diversity analysis for discrete ris-assisted systems under rank deficient channels," *IEEE Wireless Communications Letters*, 2023.
- [5] R. Xiong, X. Dong, T. Mi, K. Wan, and R. C. Qiu, "Optimal discrete beamforming of ris-aided wireless communications: An inner product maximization approach," in *2024 IEEE Wireless Communications and Networking Conference (WCNC)*, 2024, pp. 1–6.
- [6] Y. Wu, D. Xu, D. W. K. Ng, R. Schober, and W. Gerstacker, "Globally optimal resource allocation design for ris-assisted multiuser networks with discrete phase shifts," in *2023 IEEE International Conference on Communications Workshops (ICC Workshops)*, 2023, pp. 1216–1221.
- [7] X. Yu, D. Xu, and R. Schober, "Optimal beamforming for miso communications via intelligent reflecting surfaces," in *2020 IEEE 21st International Workshop on Signal Processing Advances in Wireless Communications (SPAWC)*, 2020, pp. 1–5.
- [8] M. Jung, W. Saad, M. Debbah, and C. S. Hong, "On the optimality of reconfigurable intelligent surfaces (ris): Passive beamforming, modulation, and resource allocation," *IEEE Transactions on Wireless Communications*, vol. 20, no. 7, pp. 4347–4363, 2021.
- [9] M. Abbasi Mosleh, F. Hélot, and R. Tafazolli, "Simplified capacity formulation and phase optimization in large ris-aided mimo systems," *IEEE Transactions on Wireless Communications*, vol. 24, no. 4, pp. 3362–3375, 2025.
- [10] T. Vu, R. Raich, and X. Fu, "On local linear convergence of projected gradient descent for unit-modulus least squares," *IEEE Transactions on Signal Processing*, vol. 71, pp. 3883–3897, 2023.
- [11] K. Li, C. Huang, Y. Gong, and G. Chen, "Double deep learning for joint phase-shift and beamforming based on cascaded channels in ris-assisted mimo networks," *IEEE Wireless Communications Letters*, vol. 12, no. 4, pp. 659–663, 2023.
- [12] D. Vordonis, D. Kompostiotis, and V. Paliouras, "Reconfigurable intelligent surface-aided ofdm wireless communications: Hardware aspects of reflection optimization methods," in *2022 11th International Conference on Modern Circuits and Systems Technologies (MOCAST)*. IEEE, 2022, pp. 1–4.
- [13] S. Shahabuddin, M. S. Sakib, H. Malik, and M. A. Albreem, "Self-tuning ris controller for vehicular communications: A hardware perspective," in *2025 IEEE International Symposium on Circuits and Systems (ISCAS)*, 2025, pp. 1–5.
- [14] S. Zeng, B. Di, H. Zhang, J. Gao, S. Yue, X. Hu, R. Fu, J. Zhou, X. Liu, H. Zhang, Y. Wang, S. Sun, H. Qin, X. Su, M. Wang, and L. Song, "Ris-based int-2030 testbed for mmwave multi-stream ultra-massive mimo communications," *IEEE Wireless Communications*, vol. 31, no. 3, pp. 375–382, 2024.
- [15] M. Soltanalian and P. Stoica, "Designing unimodular codes via quadratic optimization," *IEEE Transactions on Signal Processing*, vol. 62, no. 5, pp. 1221–1234, 2014.
- [16] J. Tranter, N. D. Sidiropoulos, X. Fu, and A. Swami, "Fast unit-modulus least squares with applications in beamforming," *IEEE Transactions on Signal Processing*, vol. 65, no. 11, pp. 2875–2887, 2017.
- [17] H. Guo, Y.-C. Liang, J. Chen, and E. G. Larsson, "Weighted sum-rate maximization for reconfigurable intelligent surface aided wireless networks," *IEEE Transactions on Wireless Communications*, vol. 19, no. 5, pp. 3064–3076, 2020.
- [18] T. Esmaeilbeig, K. V. Mishra, and M. Soltanalian, "Beyond diagonal ris: Key to next-generation integrated sensing and communications?" *IEEE Signal Processing Letters*, vol. 32, pp. 216–220, 2025.
- [19] A. Eamaz, F. Yeganegi, and M. Soltanalian, "Marli: Attack on the discrete-phase wsl minimization problem," *IEEE Transactions on Signal Processing*, vol. 72, pp. 219–234, 2024.
- [20] D. K. Pekcan and E. Ayanoglu, "Achieving optimum received power for discrete-phase ris with elementwise updates in the least number of steps," *IEEE Open Journal of the Communications Society*, vol. 5, pp. 2706–2722, 2024.
- [21] H. Cho, J. Moon, and J. Choi, "Low complexity ris update in ofdm systems via ris element allocation," *IEEE Transactions on Wireless Communications*, vol. 23, no. 10, pp. 15 142–15 154, 2024.
- [22] S. Kim, H. Lee, J. Cha, S.-J. Kim, J. Park, and J. Choi, "Practical channel estimation and phase shift design for intelligent reflecting surface empowered mimo systems," *IEEE Transactions on Wireless Communications*, vol. 21, no. 8, pp. 6226–6241, 2022.
- [23] Z. Wang, T. Zhou, T. Xu, Y. Zhao, and H. Hu, "Capacity characterization for reconfigurable intelligent surface aided mimo communication under practical phase shift model," in *Proceedings of the IEEE 32nd Annual International Symposium on Personal, Indoor and Mobile Radio Communications (PIMRC)*, 2021, pp. 1413–1418.
- [24] A. Sirojuddin, T. Suryani, Suwadi, Wirawan, and M.-S. Alouini, "Mimo capacity maximization with a practical model for reconfigurable intelligent surfaces," *IEEE Wireless Communications Letters*, vol. 13, no. 12, pp. 3330–3334, 2024.
- [25] E. Björnson and L. Sanguinetti, "Power scaling laws and near-field behaviors of massive mimo and intelligent reflecting surfaces," *IEEE Open Journal of the Communications Society*, vol. 1, pp. 1306–1324, 2020.
- [26] M. Journée, Y. Nesterov, P. Richtárik, and R. Sepulchre, "Generalized power method for sparse principal component analysis," *Journal of Machine Learning Research*, vol. 11, no. 2, 2010.
- [27] G. Zhou, C. Pan, H. Ren, P. Popovski, and A. L. Swindlehurst, "Channel estimation for ris-aided multiuser millimeter-wave systems," *IEEE Transactions on Signal Processing*, vol. 70, pp. 1478–1492, 2022.
- [28] S. Ragi, E. K. P. Chong, and H. D. Mittelmann, "Polynomial-time methods to solve unimodular quadratic programs with performance guarantees," *IEEE Transactions on Aerospace and Electronic Systems*, vol. 55, no. 5, pp. 2118–2127, 2019.
- [29] K. Kreutz-Delgado, "The complex gradient operator and the cr-calculus," *arXiv preprint arXiv:0906.4835*, 2009.
- [30] S. Abeywickrama, R. Zhang, Q. Wu, and C. Yuen, "Intelligent reflecting surface: Practical phase shift model and beamforming optimization," *IEEE Transactions on Communications*, vol. 68, no. 9, pp. 5849–5863, 2020.
- [31] P. Vellaisamy and A. Zeleke, "Exponential order statistics and some combinatorial identities," *Communications in Statistics - Theory and Methods*, vol. 48, no. 20, pp. 5099–5105, Oct 2019.
- [32] L. He, J. Wang, and J. Song, "Spatial modulation for more spatial multiplexing: Rf-chain-limited generalized spatial modulation aided mm-wave mimo with hybrid precoding," *IEEE Transactions on Communications*, vol. 66, no. 3, pp. 986–998, 2018.
- [33] J. Ye, S. Guo, and M.-S. Alouini, "Joint reflecting and precoding designs for ser minimization in reconfigurable intelligent surfaces assisted mimo systems," *IEEE Transactions on Wireless Communications*, vol. 19, no. 8, pp. 5561–5574, 2020.
- [34] N. C. Sagias and G. K. Karagiannidis, "Gaussian class multivariate weibull distributions: Theory and applications in fading channels," *IEEE Transactions on Information Theory*, vol. 51, no. 10, pp. 3608–3619, 2005.